AT2 Project ideas

Group projects in Assignment 2 are about taking your NLP projects to the real-world. The focus should be on **building end-user applications** that benefit user groups for specific purposes. To this end, you are encouraged to extend prior work on models that are already available open-source (if available) to build on the work of the community.

You are **free to choose the technical approaches** you would like to implement in the back end of your application - for example, exploratory data analysis, machine learning, or large language models, as long as your choice is justified in your report. Comparison between traditional and newer approaches while not mandatory to discuss in the report will be highly favoured for 'HD's.

Suggested projects are below (which indicate key problem statements, additional features are highly encouraged as relevant to the application):

1. **Personal Study Planner**

Problem statement:

Students may find it difficult to consolidate information across subjects they take and plan for their immediate tasks, but also to map their degree progression towards a career path. Imagine building your own personal study planner assistant that can help make your lives easier as a data science student.

Your task is to develop an NLP application that can help student map information from multiple sources (e.g. subject schedules) to extract key dates, deadlines and tasks to plan your study tasks in a semester. You may extract specific skills you gain from each subject, building a projected portfolio at the end of your course based on planned subjects. It can help you find blind spots such as key skills and topics currently missing in your plan, helping choose relevant electives and additional courses in the future towards a targeted job. You can be creative in the kinds of features you would like to include in the application. For instance, you can link data from job advertisement platforms to suggest relevant jobs for a skill or degree, or let the user build their portfolio by linking their github project pages etc..

Relevant Data sets/ Models: Web scraping or File uploads

2. **AI against Modern Slavery (AIMS) Explorer**

Problem statement:

Modern slavery laws generally require large companies to publish annual reports outlining their efforts to eliminate slavery from their supply chains. This transparency helps governments and citizens hold the private sector accountable and advocate for change, both locally and globally. The United Kingdom and Australia were among the first countries to adopt laws on modern slavery in 2015 and 2018, respectively. Without adequate resources, governments and NGOs struggle to properly review the volume of statements reported by organisations.

Project AIMS was initiated in 2020-2021 as part of a master's thesis by Adriana Bora (QUT). It leverages artificial (AI) techniques to help analyze corporate reporting data and promote compliance with modern slavery laws. It has since evolved into a comprehensive website of annotated datasets and research streams. Your task is to generate a web tool that can help

users navigate annotated datasets from the Australian Modern Slavery Register. The tool could support both expert users (e.g., policy analysts, legal researchers) and general users (e.g., students, activists) in understanding corporate responses to modern slavery obligations. Interactive options can include a Search & Filter Interface, Side-by-side statement comparison, trend analysis dashboard etc.

Relevant Data sets/ Models: https://github.com/mila-ai4h/ai4h_aims-au, Project AIMS (See Datasets and Tools on the website)

3. **Linguistic variation in classification tasks**

Problem statement:

Different varieties of English are often overlooked or underrepresented in NLP data sets and benchmarks, leading to text classification models not working as intended for those language variations. Your task is to build language-specific applications using data from a 3-variety benchmark for sentiment and sarcasm classification (Australian, Indian, and British varieties of English). The data set contains annotated social media posts and comments, and Google Places reviews from Australia, India and the UK. Allowing to switch between the variations either in the UI or programmatically would be a recommended feature of this application, and it can also include API integration into other systems. Additional features for aspect-based sentiment analysis, explanation of sarcasm, searchable interfaces, or explainability of models are also recommended.

Relevant Data sets/ Models: BESSTIE

4. **LLM outputs evaluation**

Problem statement:

Large Language Models (LLMs) like ChatGPT have shown impressive capabilities in generating code from natural language descriptions. However, the inherent randomness in these models can lead to variability and non-determinism in the generated outputs, raising questions about its correctness and reliability. Your task is to develop an NLP application that rigorously evaluates LLM outputs generated by comparing model outputs, considering various criteria and including iterative runs using the same/ slightly tuned prompts. The tool could support both qualitative and quantitative analysis, helping users understand how and why outputs vary, and what that means for trust and usability. It can also focus on specific high-stakes applications like education, healthcare, and software development.

Relevant Data sets/ Models: EvalPlus, Recon - multilingual evaluation benchmark

See also: EvalLM

5. **AI for information archival**

Problem statement:

Archives contain rich historical, cultural, and institutional knowledge, but much of this information remains locked in unstructured formats such as scanned documents, handwritten notes, audio files, and legacy metadata systems. Digitization efforts have made these records more available, but accessibility and discoverability remain major challenges - especially for researchers, educators, and the general public.

Your task is to build an NLP-powered application for the Australian context that enhances access to archival records using AI. The tool should support exploration, search, and interpretation of digitized archival content, making it easier to uncover connections, extract insights, and navigate large collections. You may work with digitized text, metadata, or even OCR-transcribed documents from archival repositories. Recommended features include semantic search and retrieval, historical event extraction, document summarization and visualisation.

This project aligns with ongoing research in AI for archival accessibility – see references below:

- https://link.springer.com/collections/eabccfhhbd
- http://arxiv.org/pdf/2307.14852
- https://arxiv.org/html/2501.07024v1

Relevant Data sets/ Models: Digitized archival collections (e.g., Trove, National Archives of Australia)

6. **Digitising Historical Texts for Linguistic Analysis and Language Revitalisation of Indigenous Languages in Australia**

Problem Statement

In the field of language documentation, communities working to revitalise endangered languages often rely on manually scanning historical texts and manuscripts to extract linguistic data. These materials contain valuable insights into vocabulary, grammar, and usage patterns, but the process of accessing and analysing them is currently labour-intensive and inefficient. Communities must manually scan through large volumes of text, record references in spreadsheets, and cross-reference entries to identify linguistic features such as verb usage, syntactic structures, and grammatical categories. They then need to paste these into documents to compare texts and analyse forms.

Your task is to design and develop a digital solution that transforms these historical texts into an organised, searchable application. This tool will firstly convert scanned manuscripts into searchable PDFs and from there automatically extract and categorise linguistic information. Rather than simply listing word mentions, the system will intelligently process and structure the data into meaningful linguistic sections. These may include:

- Grammatical categorisation (e.g. verbs, nouns, pronouns)
- Detection of syntactic patterns (e.g. verb-noun-case constructions)
- Identification of language conjugations and inflectional forms
- Semantic grouping (e.g. kinship terms)

Users will be able to quickly locate and analyse specific language features, such as all mentions of verbs and compare the content, thereby accelerating the reconstruction process and supporting more effective language preservation efforts and empower communities to uncover more language data. Note that this project is specifically of interest to external partners, who may be keen to offer a data science internship on the topic in 2026.

Example source text: National Library of Australia – Historical Manuscript

7. **Demographic bias auditor for text-to-image generators/ LLMs**

Problem statement:

Text-to-image generation models like DALL·E, Midjourney, and Stable Diffusion have revolutionized creative workflows, enabling users to generate realistic or artistic images from natural language prompts. However, recent research has revealed that these models often reflect and amplify demographic biases - especially in how they represent race, gender, age, and cultural identity in response to neutral or occupational prompts (e.g., "a doctor," "a teacher," "a bride", or even objects – see sample paper below).

Your task is to build an NLP+Vision application that audits demographic bias in text-to-image generation systems. The tool could allow users to input prompts and systematically analyze the generated images for patterns of bias across different demographic dimensions. This includes identifying over- or under-representation, stereotypical portrayals, and lack of diversity in generated outputs. It could also compare bias in images generated by different models, or explore language based biases only in LLMs (without image generation). You may also compare results from earlier papers to see trends in how newer models deal with bias.

Related work:

- https://dl-acm-org.ezproxy.lib.uts.edu.au/doi/pdf/10.1145/3593013.3594095
- https://proceedings.neurips.cc/paper_files/paper/2023/file/b01153e7112b347d8ed54f317840d8af-Paper-Datasets_and_Benchmarks.pdf
- https://arxiv.org/pdf/2508.03483